

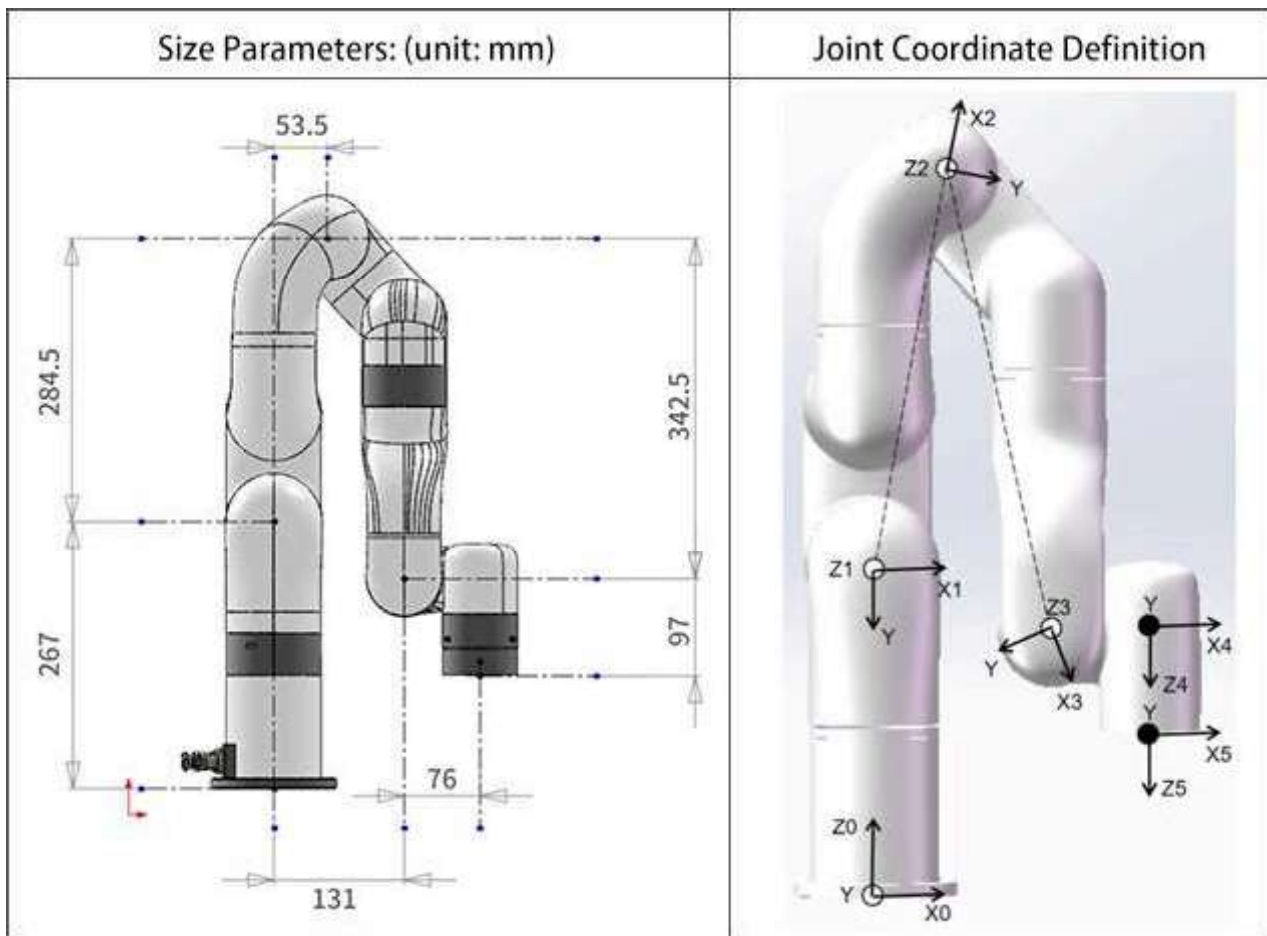
* Please note the provided Dynamic Parameters are just for reference.

1. UFACTORY xArm 5

1) UFACTORY xArm5 Modified D-H Parameters

Size Parameters: (unit: mm)			Joint Coordinate Definition		
Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	0	0	0
Joint2	0	0	$-\pi/2$	0	T2_offset
Joint3	0	0	0	a2	T3_offset
Joint4	0	0	0	a3	T4_offset
Joint5	0	97	$-\pi/2$	76	0

2) UFACTORY xArm5 Standard D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	-pi/2	0	0
Joint2	0	0	0	a2	T2_offset
Joint3	0	0	0	a3	T3_offset
Joint4	0	0	-pi/2	76	T4_offset
Joint5	0	97	0	0	0

Note:

'Tx_offset' is the offset joint angle from the mathematical zero position to the mechanical zero position shown in the picture.

$$T2_offset = -\text{atan}(284.5/53.5) = -1.3849179 \text{ } (-79.34995^\circ);$$

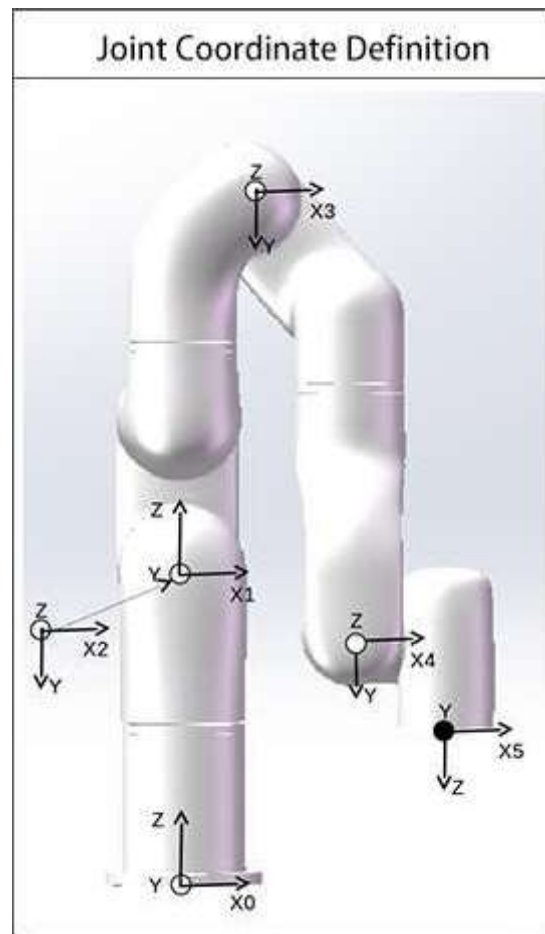
$$T3_offset = \text{atan}(284.5/53.5) + \text{atan}(0.3425/0.0775) = 2.7331843 \text{ } (156.599924^\circ);$$

$$a2 = \text{sqrt}(284.5^2 + 53.5^2) = 289.48866;$$

$$T4_offset = -\text{atan}(342.5/77.5) = -1.3482664 \text{ } (-77.249974^\circ);$$

$$a3 = \text{sqrt}(77.5^2 + 342.5^2) = 351.158796;$$

3) UFACTORY xArm5 xArm Mass Parameters



UFACTORY xArm 5-Model 1

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.177	[0.15, 27.24, -13.57]
Link2	2.011	[36.7, -220.9, 33.56]
Link3	2.01	[68.34, 223.66, 1.1]
Link4	1.206	[63.87, 29.3, 3.5]
Link5	0.17	[0, -6.77, -10.98]

UFACTORY xArm 5-Model 2

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	2.21	[38.2, -226.6, 34.7]
Link3	2.16	[69.0, 231.8, 1.0]
Link4	1.354	[65.2, 31.8, 3.11]
Link5	0.17	[0, -6.77, -10.98]

UFACTORY xArm 5-Model 3

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	2.164	[37.88, -225.4, 34.47]
Link3	2.121	[68.83, 229.85, 1.02]
Link4	1.317	[64.9, 31.2, 3.2]
Link5	0.17	[0, -6.77, -10.98]

2. UFACTORY xArm 6

1) UFACTORY xArm 6 Modified D-H Parameters